```
import torch
print(torch.cuda.is_available())
```

```
True
```

```
import pandas as pd

df = pd.read_excel("NewsSumm Dataset.xlsx")

print(df.shape)
print(df.columns)
df.head()
```

```
(348766, 6)
Index(['newspaper_name', 'published_date\n', 'headline', 'article_text',
       'human_summary', 'news_category'],
      dtype='object')
```

	newspaper_name	published_date\n	headline	article_text	human_summary	news_category
0	Indian Express	2020-06-01 00:00:00	Virus may be invisible enemy but COVID warrior...	Prime Minister Narendra Modi Monday hailed the...	Prime Minister of India said that the Virus ma...	National News
1	Economic Times	2013-02-11 00:00:00	Economy can bounce back, says PM Modi	ALLAHABAD: At least 20 persons were killed, an...	In Maha Kumbh, nearly 20 persons were killed. ...	National News
2	Business Standard	2013-02-11 00:00:00	At least 20 killed in stampede in Allahabad	At least 20 people were killed, and scores of ...	As per the sources 20 people died and scores w...	National News
3	Money Control	2013-02-11 00:00:00	Maha Kumbh: Over 20 dead in Allahabad station ...	More than 20 people were feared dead and 30 ot...	At least 20 people killed and 20 people are in...	National News
			Gandhian	This Gandhi Jayanti,	In this article, the	

```
print("Total Articles:", len(df))
```

```
Total Articles: 348766
```

```
print("Unique Sources:", df['newspaper_name'].nunique())
print(df['newspaper_name'].value_counts().head(10))
```

```
Unique Sources: 4532
newspaper_name
The Times of India    33247
The Hindu             30275
Hindustan Times       24868
Indian Express        18365
The Economic Times    15049
The Tribune           14075
The Pioneer           10610
The Indian Express     7709
Mint                   6586
The Asian Age          5946
Name: count, dtype: int64
```

```
print("Unique Categories:", df['news_category'].nunique())
print(df['news_category'].value_counts().head(10))
```

```
Unique Categories: 5491
news_category
Politics                48797
Business and Finance    33855
National News           33012
Local News              29853
International News       26443
Crime and Justice       24418
Sports                  23290
Entertainment           19898
Health and Wellness     16434
Education               13710
Name: count, dtype: int64
```

```
print(df.isnull().sum())
```

```
newspaper_name        4
published_date\n      4
headline              5
article_text         27
human_summary       240
news_category         6
dtype: int64
```

```
print(df['article_text'][0][:500])
print("\nSummary:\n", df['human_summary'][0])
```

```
Prime Minister Narendra Modi Monday hailed the contribution of medical workers during the coronavirus pande

“At the root

Summary:
 Prime Minister of India said that the Virus may be an Invisible enemy. but our warriors, medical workers a
```

```
df['article_length'] = df['article_text'].astype(str).apply(lambda x: len(x.split()))
print("Avg Article Length:", df['article_length'].mean())
```

```
Avg Article Length: 333.87469822173034
```

```
df['summary_length'] = df['human_summary'].astype(str).apply(lambda x: len(x.split()))
print("Avg Summary Length:", df['summary_length'].mean())
```

```
Avg Summary Length: 94.60826169982165
```

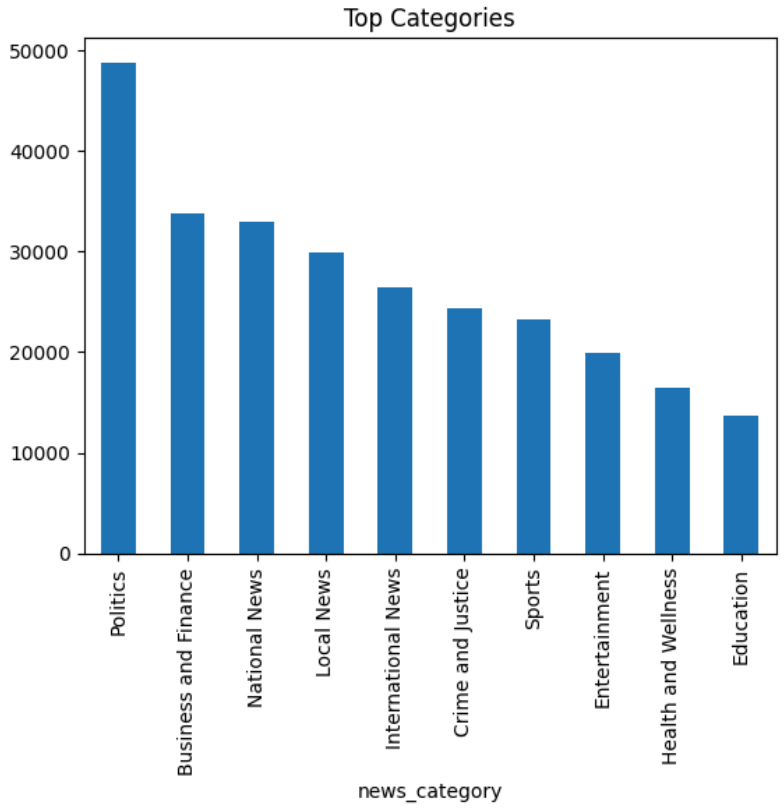
```
df['compression_ratio'] = df['article_length'] / df['summary_length']
print("Avg Compression Ratio:", df['compression_ratio'].mean())
```

```
Avg Compression Ratio: inf
```

```
from collections import Counter

words = " ".join(df['article_text'].astype(str)).split()
vocab = Counter(words)

print("Vocabulary Size:", len(vocab))
```

```
Vocabulary Size: 1918498
```

```
import matplotlib.pyplot as plt

df['news_category'].value_counts().head(10).plot(kind='bar')
plt.title("Top Categories")
plt.show()
```




```
df = df.dropna(subset=['article_text', 'human_summary', 'news_category'])
```

```
df = df.drop_duplicates(subset=['article_text'])
```

```
import re

def clean_text(text):
    text = str(text)
    text = re.sub(r'<.*?>', '', text)      # remove HTML
    text = re.sub(r'\s+', ' ', text)       # remove extra spaces
    return text.strip()

df['article_text'] = df['article_text'].apply(clean_text)
df['human_summary'] = df['human_summary'].apply(clean_text)
```

```
df['newspaper_name'] = df['newspaper_name'].str.lower().str.strip()
```

```
print(df['news_category'].value_counts().head(50))
```

```
news_category
Politics                              43398
Business and Finance                  30958
National News                         29825
Local News                            25611
International News                     24550
Crime and Justice                     22464
Sports                                21717
Entertainment                         18376
Health and Wellness                   15269
Education                             12289
Environment                            9371
Science and Technology                 8934
Lifestyle and Features                 7040
Technology and Gadgets                 6708
Opinion and Editorial                  5685
Human Interest                         5360
Miscellaneous                          4752
Religion and Spirituality              3794
Weather                                3738
Automotive                             2227
Obituaries                             1317
Politics and Government                 642
Economy                                 383
General                                 375
Crime and Legal Issues                  278
```

```
Agriculture                              206
Business and Economy                     163
economy                                  127
Infrastructure and Development           125
Infrastructure                           121
Defence                                  108
Transportation                            92
Environmental and Natural Disasters       91
Tourism                                   86
Legal                                     82
Governance                                81
Accident                                  77
History                                   72
Terrorism                                 70
Laws and Justice                          67
Industry                                  66
Other                                     63
Health and Safety                         63
Legal Affairs                             63
Accidents                                 63
International Relations                   62
Development                               61
Travel                                    57
Arts and Culture                          55
Culture                                   52
Name: count, dtype: int64
```

```
category_map = {

    # 🔷 Politics
    "Politics": "politics",
    "Politics and Government": "politics",
    "Governance": "politics",
    "Defence": "politics",
    "Terrorism": "politics",
    "International Relations": "politics",

    # 🔷 Business & Economy
    "Business and Finance": "business",
    "Business and Economy": "business",
    "Economy": "business",
    "economy": "business",
    "Industry": "business",
    "Infrastructure": "business",
    "Infrastructure and Development": "business",
    "Development": "business",

    # 🔷 General / National / Local
    "National News": "general",
    "Local News": "general",
    "General": "general",

    # 🔷 International
    "International News": "international",

    # 🔷 Crime & Law
    "Crime and Justice": "crime",
    "Crime and Legal Issues": "crime",
    "Legal": "crime",
    "Legal Affairs": "crime",
    "Laws and Justice": "crime",
    "Accident": "crime",
    "Accidents": "crime",

    # 🔷 Sports
    "Sports": "sports",

    # 🔷 Entertainment & Culture
    "Entertainment": "entertainment",
    "Arts and Culture": "entertainment",
    "Culture": "entertainment",

    # 🔷 Health
    "Health and Wellness": "health",
    "Health and Safety": "health",

    # 🔷 Education
    "Education": "education",

    # 🔷 Environment
```

```
    "Environment": "environment",
    "Environmental and Natural Disasters": "environment",

    # 🔷 Tech & Science
    "Science and Technology": "technology",
    "Technology and Gadgets": "technology",

    # 🔷 Lifestyle
    "Lifestyle and Features": "lifestyle",
    "Human Interest": "lifestyle",

    # 🔷 Religion
    "Religion and Spirituality": "religion",

    # 🔷 Weather
    "Weather": "weather",

    # 🔷 Transport & Travel
    "Automotive": "transport",
    "Transportation": "transport",
    "Travel": "transport",
    "Tourism": "transport",

    # 🔷 Agriculture
    "Agriculture": "agriculture",

    # 🔷 Obituaries
    "Obituaries": "obituaries",

    # 🔷 Opinion
    "Opinion and Editorial": "opinion",

    # 🔷 History
    "History": "history",

    # 🔷 Others
    "Miscellaneous": "other",
    "Other": "other"
}
```

```
df['broad_category'] = df['news_category'].map(category_map)

# fill missing
df['broad_category'] = df['broad_category'].fillna("other")
```

```
print(df['broad_category'].value_counts())
print("Total Broad Categories:", df['broad_category'].nunique())
```

```
broad_category
general          55811
politics         44361
business         32004
international    24550
crime            23094
sports           21717
entertainment    18483
technology       15642
health           15332
other            14880
lifestyle        12400
education        12289
environment       9462
opinion           5685
religion          3794
weather           3738
transport         2462
obituaries        1317
agriculture        206
history             72
Name: count, dtype: int64
Total Broad Categories: 20
```

```
from sklearn.feature_extraction.text import TfidfVectorizer
from sklearn.cluster import KMeans

vectorizer = TfidfVectorizer(
    stop_words='english',
    max_features=10000
)
```

```
X = vectorizer.fit_transform(df['article_text'])

kmeans = KMeans(n_clusters=25, random_state=42)

df['cluster'] = kmeans.fit_predict(X)
```

```
import numpy as np

terms = vectorizer.get_feature_names_out()

for i in range(25):
    print(f"\nCluster {i}")

    center = kmeans.cluster_centers_[i]
    top_words = center.argsort()[-10:]

    print([terms[j] for j in top_words])
```

```
Cluster 0
['security', 'terror', 'indian', 'jammu', 'army', 'pakistani', 'kashmir', 'said', 'india', 'pakistan']

Cluster 1
['government', 'sabha', 'leader', 'rahul', 'minister', 'said', 'bjp', 'gandhi', 'party', 'congress']

Cluster 2
['district', 'modi', 'prime', 'mr', 'people', 'chief', 'state', 'said', 'government', 'minister']

Cluster 3
['crop', 'farm', 'agricultural', 'state', 'river', 'agriculture', 'government', 'said', 'farmers', 'water'

Cluster 4
['character', 'like', 'cinema', 'director', 'khan', 'says', 'movie', 'actor', 'films', 'film']

Cluster 5
['mission', 'doctors', 'cancer', 'said', 'patients', 'space', 'dr', 'medical', 'hospital', 'health']

Cluster 6
['final', 'win', 'game', 'test', 'world', 'india', 'cup', 'team', 'cricket', 'match']

Cluster 7
['published', 'article', 'detailed', 'statements', 'aspects', 'developments', 'covers', 'reactions', 'quot

Cluster 8
['recorded', 'monsoon', 'normal', 'weather', 'rain', 'imd', 'celsius', 'degrees', 'rainfall', 'temperature

Cluster 9
['filed', 'order', 'judge', 'said', 'high', 'case', 'supreme', 'justice', 'bench', 'court']

Cluster 10
['flight', 'train', 'aircraft', 'traffic', 'airport', 'said', 'passengers', 'railway', 'road', 'air']

Cluster 11
['military', 'said', 'palestinians', 'war', 'palestinian', 'iran', 'hamas', 'israeli', 'gaza', 'israel']

Cluster 12
['exams', 'student', 'said', 'candidates', 'examination', 'education', 'college', 'exam', 'university', 's

Cluster 13
['trade', 'minister', 'indian', 'modi', 'russia', 'countries', 'chinese', 'said', 'china', 'india']

Cluster 14
['woman', 'allegedly', 'district', 'incident', 'station', 'case', 'accused', 'arrested', 'said', 'police']

Cluster 15
['teacher', 'learning', 'said', 'government', 'children', 'teachers', 'students', 'schools', 'education',

Cluster 16
['state', 'election', 'seats', 'assembly', 'said', 'elections', 'minister', 'congress', 'party', 'bjp']

Cluster 17
['women', 'indian', 'like', 'time', 'day', 'new', 'year', 'india', 'people', 'said']

Cluster 18
['taxes', 'government', 'said', 'rs', 'finance', 'crore', 'budget', 'income', 'gst', 'tax']
```

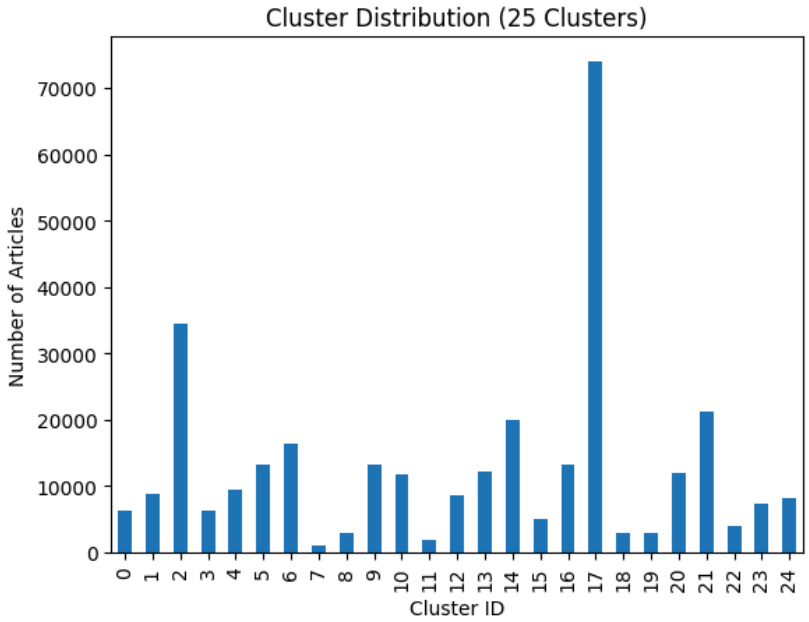
```
for i in range(5):
    print(f"\n===== Cluster {i} =====")

    sample = df[df['cluster'] == i]['headline'].head(5)
```

```
    for s in sample:
        print("-", s)
```

```
===== Cluster 0 =====
- Extradition of Laik Ali may be sought
- India, Pakistan in new Peace Talks
- Pak’s n-arsenal contains up to 90 warheads: US scientist
- ISI hand in 7/11 blasts
- Last of Pulwama attackers killed, say security forces

===== Cluster 1 =====
- Xi Jinping set to begin third term in command of China as rivals forced out
- CM busy looking after his family welfare: Bhongir MP
- IN BLACK AND WHITE Can NDA survive without Vajpayee?
- Congress dragged its feet in Bofors case: CPI
- Congress indicates movement towards formation of Telangana

===== Cluster 2 =====
- School chairman accuses former Minister of ₹700 crore fraud
- British MPs finally approve Brexit deal
- Economy can bounce back, says PM Modi
- Forty Years Ago: Parliamentary System
- Fire at Indonesian oil depot kills 17, thousands evacuated  Read more at: http://timesofindia.indiatimes.

===== Cluster 3 =====
- Do collapsed wells portend tremors in Kerala?
- Farmers unhappy with new land acquisition rates
- Pump set motors to be replaced over five years
- Tata commissions SlS Plant in Joda
- Yatra for Preservation of water

===== Cluster 4 =====
- Disha’s set for her second movie with Sanchari Vijay  Read more at: http://timesofindia.indiatimes.com/ar
- Doctor G
- Who’s the ghost in Bhool Bhulaiyaa 2??
- In memory of the late actor Jagdeep Jaffrey
- Title track of Love You Rachchu to be released on December 1
```

```
cluster_names = {
    0: "Politics",
    1: "Sports",
    2: "Business",
    3: "Entertainment"
}

df['cluster_name'] = df['cluster'].map(cluster_names)
```

```
import matplotlib.pyplot as plt

df['cluster'].value_counts().sort_index().plot(kind='bar')

plt.xlabel("Cluster ID")
plt.ylabel("Number of Articles")
plt.title("Cluster Distribution (25 Clusters)")

plt.show()
```




```
import pandas as pd

cross_tab = pd.crosstab(df['broad_category'], df['cluster'])

print(cross_tab)
```

```
cluster           0     1      2     3     4     5      6   7     8     9   \
broad_category
agriculture       0     0     13   121     0     3      0   0     0     0
business         51    51   1654   296    89   201     59  58    13   350
crime           457   107   1641    96    75   216     24  50     5  4779
education        14    20    695    66    45   249     19  47    10   199
entertainment    33    17    158    49  7506   122    163  46    17   133
environment      10    19    994  1103   114   241     12  43   479   157
general        1520  1037  10167  1937   476  1400    455  95   373  3071
health           10    24    631   229    38  6252     23  47    36   136
history           5     2      3     0     1     0      0   0     0     1
international  2308    84   1458   135   159   452    326  51    20   543
lifestyle        54    33    736   224   299   865     84  95    22   209
obituaries       17    28    151     7    60    42     29  53     0    22
opinion         144   132    686   109   110   159     70  45    11   214
other           452   422   2055   384   168   371    175  51    46   687
politics        929  6718  11712  1030   140   305    103  53    27  2189
religion         26    28    553    26    30    57     12  60     1   219
sports          260    16    555    19    50    68  14759  45     5   128
technology       47    26    415   257    59  2236    105  87    51   134
transport         2     4     85     8     8    16     16  43     1    13
weather          15     9    178   274     7    57     10  29  1868    10

cluster         ...    15     16     17    18    19    20    21   22    23  \
broad_category  ...
agriculture     ...     0      0     32     0     0    11     9    1     1
business        ...    50     73   3673  1550  1019  7732  7982  758    87
crime           ...   164    167   3536   101    20    84   273   19   137
education       ...  2993     22   1837    11     2    66   416   30    52
entertainment   ...    50     63   9038    25    13    27   317   26    41
environment     ...    46     14   3893     8    64    65   148  902    43
general         ...   813   1474  12793   530   205  1178  1539  667  1482
health          ...   119     29   2434    13    16   134   310   21  3944
history         ...     1      0     53     0     0     0     0    0     0
international   ...    98    216   6751   105    74   727   951  311   429
lifestyle       ...   134     57   7341    38    45   177   504   65   159
obituaries      ...    14     17    533     3     4     8    31    2    21
opinion         ...   100    207   2066    74    22   262   358   93    79
other           ...   116    588   4056   185    98   758   800  230   187
politics        ...   200  10161   4796   317    34   409   416  239   263
religion        ...    16     63   2374     4     4    15    23    4    20
sports          ...    72     23   4949    16    18    19   118    9    50
technology      ...    90     19   2780    43   303   176  6745  401   234
transport       ...     4      2    400     7  1064    29   180   11     6
weather         ...    18     12    654     2     5    19    20  105    13

cluster           24
broad_category
agriculture       11
business        4156
crime            256
```

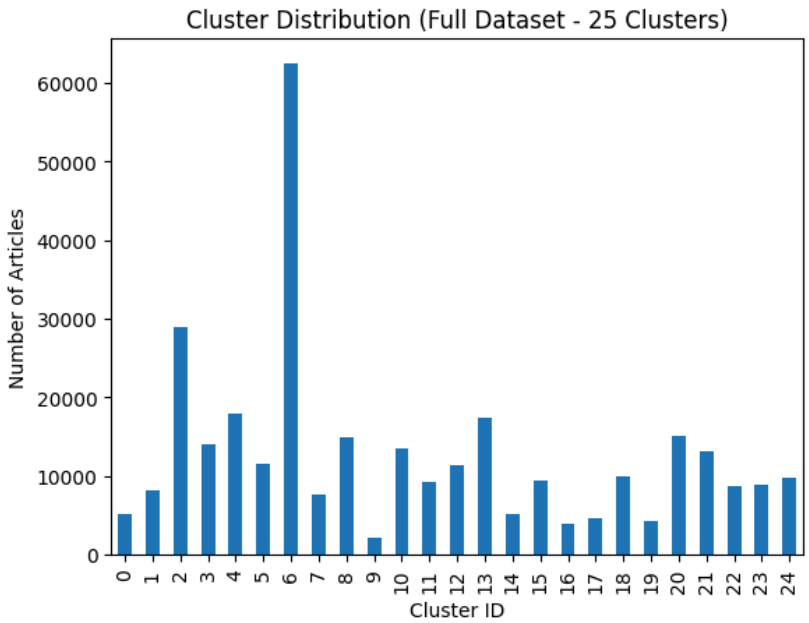
```
education        103
entertainment    210
environment       57
general         1451
health           103
history            0
international      82
```

```
from sklearn.feature_extraction.text import TfidfVectorizer

vectorizer = TfidfVectorizer(
    stop_words='english',
    max_features=5000,      # reduce memory
    max_df=0.8,             # ignore very common words
    min_df=5                # ignore rare words
)

X = vectorizer.fit_transform(df['article_text'])
```

```
from sklearn.cluster import MiniBatchKMeans

kmeans = MiniBatchKMeans(
    n_clusters=25,
    batch_size=10000,
    random_state=42
)

df['cluster'] = kmeans.fit_predict(X)
```

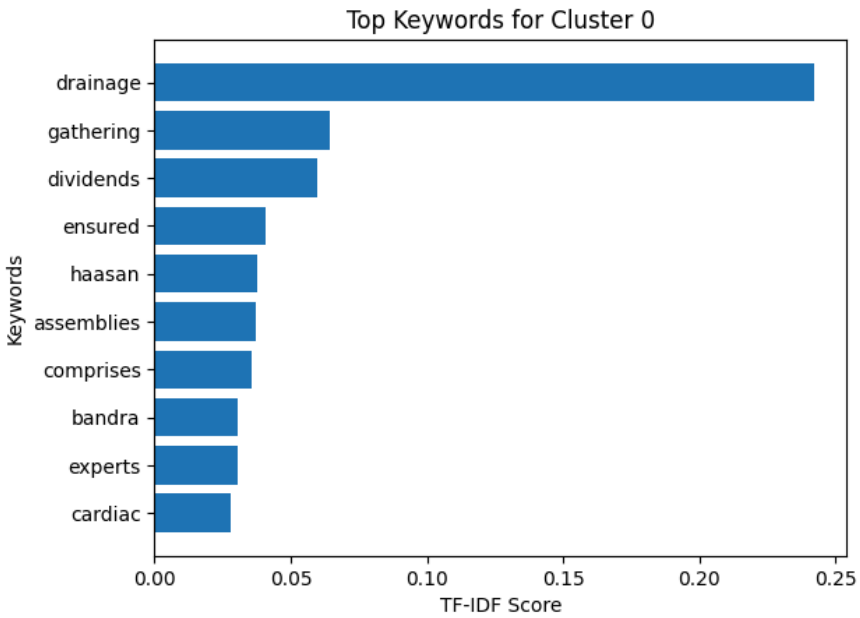
```
df.to_csv("clustered_dataset_25.csv", index=False)
```

```
import matplotlib.pyplot as plt

df['cluster'].value_counts().sort_index().plot(kind='bar')

plt.xlabel("Cluster ID")
plt.ylabel("Number of Articles")
plt.title("Cluster Distribution (Full Dataset - 25 Clusters)")

plt.show()
```




```
import matplotlib.pyplot as plt

# Example: visualize one cluster
cluster_id = 0

center = kmeans.cluster_centers_[cluster_id]
top_indices = center.argsort()[-10:]

top_words = [terms[i] for i in top_indices]
top_scores = center[top_indices]
```

```
plt.figure()
plt.barh(top_words, top_scores)

plt.xlabel("TF-IDF Score")
plt.ylabel("Keywords")
plt.title(f"Top Keywords for Cluster {cluster_id}")

plt.show()
```




```
import pandas as pd
import matplotlib.pyplot as plt

cross_tab = pd.crosstab(df['broad_category'], df['cluster'])
```